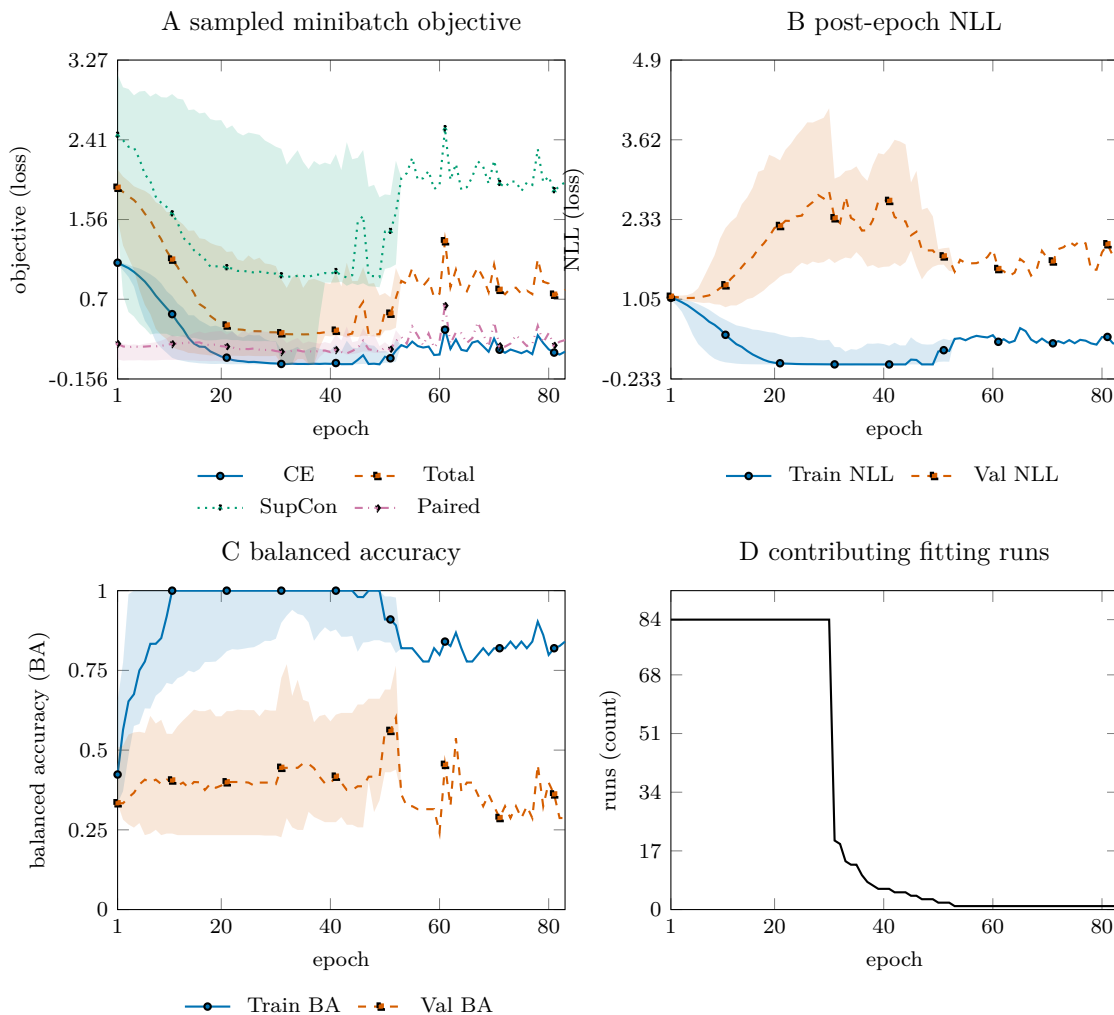


planned=87, monitored=84, missing-history=3, status=partial\_history\_coverage



RQ-S01 / Scope E: descriptive training diagnostics. 400-1800 cm<sup>-1</sup> inputs: SG uses impulse replacement and Savitzky-Golay smoothing; arPLS uses impulse replacement and baseline correction; both finish with [0,1] min-max scaling. CE = chemical cross-entropy; SupCon = supervised contrastive; Paired = paired consistency; NLL = negative log-likelihood; BA = balanced accuracy. Authenticated complete monitor histories aggregated across unique fitting jobs into descriptive policy/recipe/stage curves. Curves summarize per-epoch minibatch-weighted training chemical cross-entropy and the composite total optimization objective; these sampled minibatch objectives differ from post-epoch evaluation negative log-likelihood. Auxiliary supcon/paired loss components are recipe-specific and are not directly comparable to the total objective across recipes. Validation metrics are recorded for source\_fit runs only and never substitute for a held-out test set; calibration\_model\_fit and final\_refit rows carry training-only diagnostics. Source\_fit optimization uses a 30-epoch minimum stopping threshold and a 200-epoch maximum; refit stages reuse the caller-selected duration. Curves are fitting-run-weighted descriptive summaries over correlated runs (seeds/splits). Lines show medians; 10th/90th percentiles describe variation among saved fitting runs, not confidence intervals. Only runs that reached each epoch contribute; early stopping changes the composition, and stopped runs are not carried forward. Fitting jobs are not independent chemical samples: the dataset has 598 spectra from 69 physical samples and 10 instruments. Coverage is limited to saved SG/arPLS monitor histories; historical MIN losses are not pooled or invented. A missing history is not a zero loss or proof of a failed model. These curves do not support hypothesis tests, superiority claims, model selection, or new uncertainty inference. P08 learned-method performance is reported elsewhere.